Light

# GCSE Computer Science — Embedded Systems (Set 2)

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GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A coffee machine in an office allows users to select different drinks by pressing buttons. A small computer inside controls the brewing process. What type of computer system is built into this coffee machine?

- ○ **A.** An embedded system
- ○ **B.** A mainframe computer
- ○ **C.** A desktop computer
- ○ **D.** A server computer

Q2 (1 mark)

Many devices in everyday life contain embedded systems.
Which of the following contains an embedded system?

- ○ **A.** A laptop running a web browser
- ○ **B.** A games console running downloaded games
- ○ **C.** A smoke detector that sounds an alarm when smoke is detected
- ○ **D.** A smartphone running multiple applications

Q3 (1 mark)

A lift (elevator) in an office building is controlled by a computer system. Which characteristic makes this an embedded system?

- ○ **A.** It can browse the internet to check weather forecasts
- ○ **B.** It can run any software the building manager installs
- ○ **C.** It is dedicated to controlling lift operations and cannot run other software
- ○ **D.** It requires a keyboard and mouse for operation

Q4 (1 mark)

An automatic security light turns on when someone walks past it at night.
What type of component detects the person's movement?

- ○ **A.** An actuator
- ○ **B.** A processor
- ○ **C.** A display
- ○ **D.** A sensor

Q5 (1 mark)

A vending machine dispenses a drink when a customer makes a selection and pays.
What type of component physically releases the drink from the machine?

- ○ **A.** A sensor
- ○ **B.** A processor
- ○ **C.** An actuator
- ○ **D.** A memory chip

Q6 (1 mark)

A company designs a digital bathroom scale that displays weight.
Which is an advantage of using an embedded system in this scale?

- ○ **A.** The embedded system is small, low-cost, and uses minimal power
- ○ **B.** The scale can run spreadsheet software to track weight over time
- ○ **C.** Users can install apps to add new features to the scale
- ○ **D.** The scale requires an internet connection to function

Q7 (1 mark)

A car's speedometer must display the current speed without any noticeable delay as the car accelerates or brakes.
What characteristic of embedded systems makes this instant response possible?

- ○ **A.** Real-time processing that responds immediately to sensor inputs
- ○ **B.** The ability to connect to satellite navigation systems
- ○ **C.** The ability to download software updates while driving
- ○ **D.** Large storage capacity for recording journey history

Q8 (1 mark)

A burglar alarm system monitors sensors around a house and sounds an alarm if an intruder is detected.
What type of computer system controls the burglar alarm?

- ○ **A.** A personal computer
- ○ **B.** A tablet computer
- ○ **C.** A supercomputer
- ○ **D.** An embedded system

Q9 (1 mark)

A student is explaining the difference between embedded systems and general-purpose computers to a friend.
Which statement correctly describes embedded systems?

- ○ **A.** Embedded systems can run any software like a laptop
- ○ **B.** Embedded systems are always connected to the internet
- ○ **C.** Embedded systems perform specific, dedicated tasks
- ○ **D.** Embedded systems always have screens and keyboards

Q10 (1 mark)

The software that runs on an embedded system is often called firmware. Which statement correctly describes firmware?

- ○ **A.** Firmware is stored permanently in the device and controls its dedicated functions
- ○ **B.** Firmware is downloaded from the internet each time the device turns on
- ○ **C.** Firmware allows users to install new applications whenever they want
- ○ **D.** Firmware must be updated daily to keep the device working

Q11 (1 mark)

A wireless doorbell button operates for over a year on a small battery. Which characteristic of embedded systems enables this long battery life?

- ○ **A.** High processing power for complex calculations
- ○ **B.** Large memory capacity for storing data
- ○ **C.** Low power consumption due to minimal components
- ○ **D.** Constant internet connectivity for updates

Q12 (1 mark)

A medical insulin pump delivers precise doses of insulin to a diabetic patient throughout the day.
Which characteristic of embedded systems is most important for this medical device?

- ○ **A.** Ability to play music to entertain the patient
- ○ **B.** Ability to run different medical applications
- ○ **C.** Large screen for watching videos
- ○ **D.** High reliability for continuous, accurate operation

Q13 (2 marks)

An automatic hand sanitiser dispenser releases sanitiser when hands are placed underneath it.
Which sequence correctly describes how this embedded system operates?

- ○ **A.** User presses button → Display shows message → Sanitiser is released
- ○ **B.** Timer counts down → Alarm sounds → User collects sanitiser
- ○ **C.** Actuator detects hands → Sensor processes signal → Processor releases sanitiser
- ○ **D.** Sensor detects hands → Processor activates pump → Actuator dispenses sanitiser

Q14 (2 marks)

A company is designing a digital kitchen timer that counts down and beeps when time is up.
Which combination of factors correctly explains why an embedded system is suitable?

- ○ **A.** Low cost, simple dedicated function, and long battery life
- ○ **B.** High processing power, large display, and internet connectivity

- ○ **C.** Ability to run apps, touchscreen interface, and voice control
- ○ **D.** Large storage, multiple user accounts, and wireless printing

Q15 (2 marks)

A factory conveyor belt system must stop within 100 milliseconds when a safety sensor detects an obstruction.
Which statement correctly explains why this is a real-time system requirement?

- ○ **A.** Because the conveyor belt moves very fast and looks impressive in real time
- ○ **B.** Because missing the 100ms deadline could result in injury or damage, making guaranteed response essential
- ○ **C.** Because the system displays belt speed in real time on a monitor
- ○ **D.** Because workers need to see conveyor status updated every 100 milliseconds

Q16 (2 marks)

A company sells thousands of smart doorbells containing embedded systems. Later, they discover the video quality could be improved with a software update.
Which statement correctly identifies a disadvantage of embedded systems in this situation?

- ○ **A.** The doorbells cannot record video because embedded systems do not support cameras
- ○ **B.** Delivering software updates to embedded devices is challenging, and some customers may never update their devices
- ○ **C.** The embedded systems will automatically update themselves

overnight without user involvement

○ **D.** Embedded systems can easily be upgraded by replacing the processor chip

Q17 (2 marks)

A smart garden irrigation system waters plants based on environmental conditions.
Which combination of sensors would help the system decide when to water?

○ **A.** Fingerprint sensor and barcode scanner

○ **B.** Soil moisture sensor and rain sensor

○ **C.** Keyboard input and mouse clicks

○ **D.** Microphone and speaker

Q18 (2 marks)

A train's automatic door system opens doors only when the train has stopped at a platform.
Which statement correctly explains why an embedded system is preferred for this application?

○ **A.** Embedded systems allow passengers to control doors using smartphone apps

○ **B.** Embedded systems allow train drivers to install custom door-opening software

○ **C.** Embedded systems can display train timetables and entertainment during journeys

○ **D.** Embedded systems provide reliable, dedicated control that operates consistently without the unpredictability of general-purpose

systems

Q19 (2 marks)

A company is designing an embedded system for a children's electronic toy.
Which design constraints must the engineers consider that differ from designing desktop software?

- ○ **A.** The toy must have limited memory, low power consumption, and fit within a small physical space
- ○ **B.** The toy must run the latest version of Windows and support Microsoft Office
- ○ **C.** The toy must connect to a 4K monitor and support high-definition gaming
- ○ **D.** The toy must have at least 1TB of storage and 32GB of RAM

Q20 (2 marks)

A smart thermostat maintains a room at the temperature set by the user. Which description correctly explains the control process used by this embedded system?

- ○ **A.** The system heats continuously until the user manually turns it off
- ○ **B.** The system only works when connected to the internet to receive temperature data
- ○ **C.** The system predicts temperature based on weather forecasts and heats in advance
- ○ **D.** The system uses a feedback loop: measuring temperature, comparing to the set point, and adjusting heating or cooling

Q21 (2 marks)

A car's engine management system stores its control program in flash memory rather than a hard disk drive.
Which statement correctly explains this design choice?

- ○ **A.** Flash memory can store more data than hard drives in vehicles
- ○ **B.** Flash memory has no moving parts, making it reliable in the vibrating environment of a car engine
- ○ **C.** Flash memory requires an internet connection to function properly
- ○ **D.** Flash memory allows drivers to install their own engine control software

Q22 (2 marks)

Modern homes contain many embedded systems.
Which list contains only examples of household embedded systems?

- ○ **A.** Laptop, tablet, and smartphone
- ○ **B.** Microwave timer, washing machine controller, and thermostat
- ○ **C.** Desktop computer, games console, and smart TV apps
- ○ **D.** Web browser, word processor, and spreadsheet software

Q23 (2 marks)

A company is developing an embedded system for a new electronic blood pressure monitor.
Why must the software be more thoroughly tested than typical desktop applications?

- ○ **A.** Because desktop applications are never tested before release
- ○ **B.** Because errors in medical devices could harm patients, and updates after sale are difficult
- ○ **C.** Because embedded systems always have more lines of code than desktop software
- ○ **D.** Because desktop applications can automatically fix their own bugs

Q24 (3 marks)

An aircraft manufacturer is deciding between using a general-purpose computer running Linux or a dedicated embedded system for the autopilot.
The autopilot must:
• Respond to sensor inputs within 10 milliseconds

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• Operate for 20+ hours continuously without failure
• Meet aviation safety certification requirements
• Function reliably at altitude with temperature variations
```

Which statement best evaluates these options?

- ○ **A.** Linux is better because it is widely used and has extensive community support for troubleshooting
- ○ **B.** A dedicated embedded system is better because it provides deterministic real-time response, certified reliability, and resilience to environmental conditions
- ○ **C.** Linux is better because pilots can install additional navigation apps during flight
- ○ **D.** Both options are equally suitable because modern computers are all reliable enough for aviation

Q25 (3 marks)

An automated car wash system uses an embedded system to control the washing process.
The system includes:
• Vehicle position sensors
• Water spray nozzles
• Rotating brushes
• Dryer fans
• Payment terminal
• Display showing wash progress
A student classifies these as inputs or outputs:
1. Vehicle position sensors → INPUT

```
2. Water spray nozzles → INPUT
3. Rotating brushes → OUTPUT
4. Dryer fans → OUTPUT
5. Payment terminal → OUTPUT
6. Display → OUTPUT
```

Which rows contain errors?

- ○ **A.** Rows 1 and 3 only
- ○ **B.** Rows 2 and 5 only
- ○ **C.** Rows 3 and 6 only
- ○ **D.** Rows 4 and 5 only

Q26 (3 marks)

A nuclear power plant uses embedded systems to monitor reactor temperature and control cooling systems.
The system must:
• Read temperature sensors every 50 milliseconds

```
• Activate emergency cooling within 200 milliseconds of detecting
• Log all readings to a database for later analysis
```

An engineer proposes using a standard Windows PC for all three functions.
Which statement correctly evaluates this proposal?

- ○ **A.** The proposal is acceptable because Windows PCs can process data faster than embedded systems
- ○ **B.** The proposal is acceptable for logging but unacceptable for safety-critical monitoring and control because Windows cannot guarantee real-time response
- ○ **C.** The proposal is unacceptable for all functions because Windows cannot read temperature sensors
- ○ **D.** The proposal is acceptable because Windows 11 includes real-time features for industrial control

Q27 (3 marks)

A company is designing a smart electricity meter for homes. The meter must:
• Measure electricity usage continuously

```
• Display current usage to the homeowner
• Transmit usage data to the electricity company
• Operate reliably for 15+ years
```

The design team debates two approaches:
• Option A: A single embedded processor handling all functions
• Option B: Separate embedded processors for metering, display, and communication
Which statement correctly evaluates these design approaches?

- ○ **A.** Option A is always better because fewer processors means lower cost and simpler design
- ○ **B.** Option B is better because isolating metering from communication ensures billing accuracy even if the communication system fails or is compromised

○ **C.** Option A is better because multiple processors would require the meter to be larger than a house

○ **D.** Both options are identical in reliability and security because all processors are embedded

Q28 (3 marks)

A student makes the following statements comparing embedded system development to desktop software development:
1. Embedded developers must consider hardware limitations like limited RAM
2. Embedded software can always be updated over the internet like smartphone apps
3. Embedded developers must often write code that interacts directly with hardware
4. Embedded systems have unlimited processing power because chips are cheap
Which statements are correct?

○ **A.** Statements 1 and 3 only are correct

○ **B.** Statements 2 and 4 only are correct

○ **C.** Statements 1, 2, and 3 only are correct

○ **D.** All statements are correct

Q29 (3 marks)

A city installs thousands of smart streetlights with embedded systems that:
• Adjust brightness based on ambient light
• Report faults to a central maintenance system
• Can be remotely dimmed to save energy
A security analyst warns about potential vulnerabilities.
Which statement correctly identifies security challenges specific to these embedded systems?

○ **A.** Streetlights cannot be attacked because they do not store valuable personal data

○ **B.** Security is not a concern because embedded systems are too simple to hack

○ **C.** The network connectivity creates attack surfaces, firmware updates across thousands of devices are difficult, and compromised lights could be used to attack other systems or cause traffic accidents

○ **D.** The streetlights are completely secure because they use dedicated embedded systems rather than general-purpose computers

Q30 (3 marks)

A manufacturer is choosing between an embedded system and a general-purpose computer for a factory robot that performs precision welding 24/7.
Consider these factors:
• Timing precision for weld quality
• Continuous operation requirements
• Cost of system failure
• Need for software flexibility
Which statement correctly evaluates the choice?

○ **A.** A general-purpose computer is better because factory robots need maximum processing power for complex calculations

○ **B.** An embedded system is better for timing precision and reliability, but if the welding patterns change frequently, the difficulty of updating embedded software becomes a significant disadvantage

○ **C.** A general-purpose computer is better because it can be easily reprogrammed when welding requirements change

○ **D.** Both systems would provide identical performance because welding is a simple repetitive task

Total: **55** marks • Time: **30 min**

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